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AI 소프트웨어학과

머신러닝1

- 추론통계 -
분산의 분포

1. Life_case.csv 데이터에서 Case에 대해 Person_A, B의 관련이 있는지 아닌지 판단하고, 관련이 있다면 Case 변수들 중 Person_A, B의 횟수가 가장 비슷한 Case 2개를 추출하시오.(p-value : 0.05)

```
data=read.csv("Life_case_1번문제.csv", row.names = 1)
```

```
chi_square_test_result <- chisq.test(data)
```

```
print(chi_square_test_result)
```

Pearson's Chi-squared test

```
data: data
```

```
X-squared = 24.378, df = 4, p-value = 6.709e-05
```

1. Life_case.csv 데이터에서 Case에 대해 Person_A, B이 관련이 있는지 아닌지 판단하고, 관련이 있다면 Case 변수들 중 Person_A, B의 횃수가 가장 비슷한 Case 2개를 추출하시오.(p-value : 0.05)

```
results <- chisq.posthoc.test(data, method = "bonferroni")
```

```
results
```

	Dimension	Value	Person_A	Person_B
1	Elderly parent care	Residuals	-1.5249754	1.5249754
2	Elderly parent care	p values	1.0000000	1.0000000
3	Small online shop	Residuals	3.4428467	-3.4428467
4	Small online shop	p values	0.0057560	0.0057560
5	Apartment community	Residuals	-3.9291150	3.9291150
6	Apartment community	p values	0.0008530	0.0008530
7	Pet care	Residuals	0.6128870	-0.6128870
8	Pet care	p values	1.0000000	1.0000000
9	shared travel planning	Residuals	0.6402743	-0.6402743
10	shared travel planning	p values	1.0000000	1.0000000

2. Life_case.csv 데이터에서 Activity가 Person_A, B에 독립인지 아닌지를 판단하고, 만약 독립이라면 Person A, B의 활동을 비교했을 때, A, B의 활동의 차이가 가장 큰 Top 5를 도출하시오.(p-value : 0.05)

```
Pearson's Chi-squared test
```

```
data: data
```

```
X-squared = 820.71, df = 39, p-value < 2.2e-16
```

2. Life_case.csv 데이터에서 Activity가 Person_A, B에 독립인지 아닌지를 판단하고, 만약 독립이라면 Person A, B의 활동을 비교했을 때, A, B의 활동의 차이가 가장 큰 Top 5를 도출하시오.(p-value : 0.05)

	Dimension	Value	Person_A	Person_B
1	Hospital appointment booking	Residuals	-3.907454	3.907454
2	Hospital appointment booking	p values	0.007462	0.007462
3	Medication refill tracking	Residuals	4.952541	-4.952541
4	Medication refill tracking	p values	0.000059	0.000059
5	Insurance claim paperwork	Residuals	-4.322007	4.322007
6	Insurance claim paperwork	p values	0.001237	0.001237
7	weekly grocery delivery setup	Residuals	4.550275	-4.550275
8	weekly grocery delivery setup	p values	0.000429	0.000429
9	calling clinic or pharmacy	Residuals	-2.080697	2.080697
10	calling clinic or pharmacy	p values	1.000000	1.000000
11	Transportation to appointments	Residuals	-6.698593	6.698593
12	Transportation to appointments	p values	0.000000	0.000000
13	Meal prep for parent	Residuals	6.901639	-6.901639

3. 카페 손님들의 만족도에 좋은 영향을 미치는 각변수별 항목들은 무엇인가?(p-value : 0.05)

```
read.csv("cafe.csv", stringsAsFactors = TRUE)
```

```
#분산분석
```

```
anova_result <- aov(Satisfaction ~ SeatLocation + CafeMood + CoffeeType, data = data)
```

```
summary(anova_result)
```

```

              Df Sum Sq Mean Sq F value    Pr(>F)
data$SeatLocation    2   666.2    333.1    13.51 5.50e-06 ***
data$CafeMood         2 2570.5   1285.3    52.11 < 2e-16 ***
data$CoffeeType       2   861.7    430.9    17.47 2.45e-07 ***
Residuals          113 2786.9     24.7
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
> leveneTest(Satisfaction ~ SeatLocation, data = data)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	1.024	0.3623
	117		

```
> leveneTest(Satisfaction ~ CafeMood, data = data)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	0.26	0.7715
	117		

```
> leveneTest(Satisfaction ~ CoffeeType, data = data)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	0.6328	0.5329
	117		

```
tukey_result <- glht(anova_result, linfct = mcp(SeatLocation = "Tukey",
CafeMood = "Tukey",
CoffeeType = "Tukey"))
```

```
summary(tukey_result)
```

	Estimate	Std. Error	t value	Pr(> t)
SeatLocation: Terrace - Center == 0	0.3786	1.1856	0.319	0.9998
SeatLocation: Window - Center == 0	3.2160	1.1527	2.790	0.0493 *
SeatLocation: Window - Terrace == 0	2.8373	1.0869	2.610	0.0793 .
CafeMood: Music - Crowded == 0	7.8018	1.1285	6.913	<1e-04 ***
CafeMood: Quiet - Crowded == 0	10.9913	1.0915	10.070	<1e-04 ***
CafeMood: Quiet - Music == 0	3.1895	1.1645	2.739	0.0566 .
CoffeeType: Espresso - Americano == 0	-3.1331	1.1091	-2.825	0.0448 *
CoffeeType: Latte - Americano == 0	3.5942	1.1269	3.190	0.0154 *
CoffeeType: Latte - Espresso == 0	6.7273	1.1384	5.910	<1e-04 ***

```
---
```

```
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
```